AI-Based Smart Streetlight Energy Optimization

Capstone Project – Second-Year EEE

1. Abstract

This project demonstrates an educational smart streetlight prototype that uses ambient-light and motion inputs to select an appropriate lighting mode. A Random Forest classifier predicts OFF, DIM, or HIGH mode from simulated sensor data. A Streamlit dashboard displays predictions, model metrics and assumption-based energy estimates.

2. Problem Statement

Streetlights can waste energy when operated at full brightness regardless of ambient light or movement. Adaptive lighting can reduce unnecessary usage while responding to motion in dark conditions.

3. Objectives

- Simulate timestamp, ambient-light and motion data.
- Preprocess data and explore sensor patterns.
- Train and evaluate a classification model.
- Predict OFF, DIM or HIGH lighting mode.
- Estimate energy consumption and show a dashboard.

4. Tools and Technologies

Python, Pandas, NumPy, scikit-learn, Matplotlib and Streamlit. Optional hardware: ESP32/Arduino, LDR module, PIR sensor, low-voltage LED and a correctly rated driver.

5. Methodology

Generate simulated data; assign labels using a stated control policy; split the data into training and test sets; train a Random Forest classifier; evaluate accuracy, precision, recall, F1-score and confusion matrix; test sensor values in the dashboard; compare energy use under documented assumptions.

6. System Block Diagram

LDR + PIR + Time → Data preprocessing → Random Forest classifier → OFF / DIM / HIGH decision → LED driver → Low-voltage LED

7. Control Policy

OFF when ambient light is sufficient (≥250 lux); HIGH when dark and motion is detected; DIM when dark and no motion is detected. Demonstration brightness levels are 0%, 100% and 35%, respectively.

8. Energy Calculation

Assumptions: one 30 W LED, 12 hours per night; smart policy uses HIGH mode for 25% of the night and DIM mode at 12 W for 75%.

Baseline = 30 × 12 / 1000 = 0.360 kWh/night.

Smart policy = (30 × 0.25 × 12 + 12 × 0.75 × 12) / 1000 = 0.198 kWh/night.

Estimated saving = (0.360 – 0.198) / 0.360 × 100 = 45%. This is not measured field performance.

9. Results and Limitations

The software displays a predicted lighting mode, model evaluation metrics and energy comparison. Since the dataset and labels are simulated, scores mainly demonstrate the workflow and do not prove performance in real streets. Real sensor collection and field validation are required.

10. Safety

For hardware demonstrations, use a low-voltage supply and a correctly rated LED driver. Never connect an LED directly to a GPIO pin. Do not work with mains voltage; have a qualified instructor check the setup.

11. Conclusion

The project illustrates how sensor inputs and machine learning can support adaptive streetlight control. Future work includes collecting real sensor data, testing under varied conditions, calibrating sensors and validating actual energy savings.

12. Viva Questions

What is an LDR? A light-dependent resistor whose resistance varies with light intensity.

What is a PIR sensor? A sensor that detects motion-related changes in infrared radiation.

Why use Random Forest? It combines decision trees and can learn non-linear patterns.

What are the output classes? OFF, DIM and HIGH.

How is energy calculated? Power in watts multiplied by hours, divided by 1000.

Is 45% saving measured? No; it is an illustrative estimate based on assumptions.